Big Data Project

Knowledge Base Enrichment & Dimensionality Reduction

Cross-Platform Music Analytics — Spotify & Apple Music

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228 Enriched Tracks

5 Acoustic Clusters

3 Dim. Reduction Methods

Outine



01

Methodology & Data Enrichment

Datasets, fusion strategy, standardization



02

Exploratory Data Analysis

Distributions, correlations, outliers



03

K-Means Clustering

Elbow method, 5 cluster profiles



04

DBSCAN

K-distance graph, comparison with K-Means



05

Dimensionality Reduction

PCA · t-SNE · ISOMAP · Evaluation



06

Conclusion

Key findings & future directions

01 — Methodology & Data Enrichment



Dataset 1

Spotify API
Metadata + Popularity

Dataset 2

Acoustic Signal
Features

Text Norm.

clean_text()
Lowercase · Regex

Inner Join

228 unique tracks

clean_text() Function

- Convert to lowercase
- Remove text inside ()
- Strip feat. & remix info
- Eliminate special chars

Features Retained

- Popularity (outcome only)
- Danceability · Energy · Valence
- Acousticness · Liveness · Tempo
- Duration · Loudness · Speechiness

02 — Exploratory Data Analysis

228	−0.12	+0.76	−0.71
Merged Tracks	Danceability ↔ Popularity (r)	Energy ↔ Loudness (r)	Energy ↔ Acousticness (r)

Multi-modal Popularity

Distribution shows 3 peaks: niche [0–10], mid-tier [10–20], mainstream [20–90]. Not a normal distribution — clustering is justified.

Acousticness Right-Skewed

Most tracks score below 0.3, confirming the dataset is dominated by electronically produced music (pop, hip-hop, electronic).

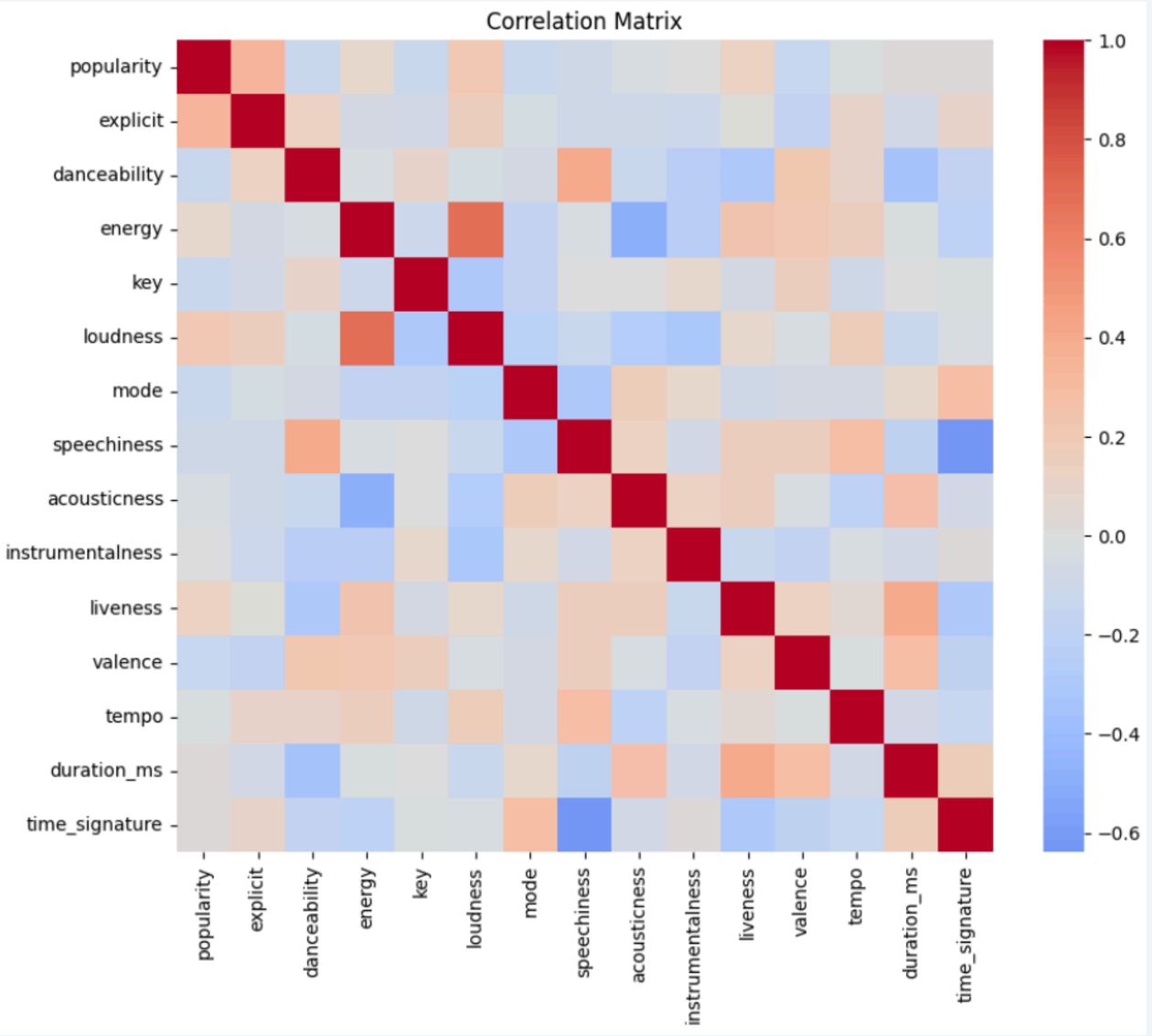
Popularity is Multi-Causal

No single acoustic feature correlates strongly with popularity (|r| < 0.15 for all). Commercial success depends on non-acoustic factors.

02 — Correlation Matrix Highlights

Feature Pair	Pearson r	Interpretation
Energy ↔ Loudness	**+0.76**	Strong positive — louder tracks are more energetic (production-driven)
Energy ↔ Acousticness	**−0.71**	Strong negative — electronic production reduces acousticness
Valence ↔ Danceability	**+0.38**	Moderate — happy-sounding tracks tend to be more danceable
Danceability ↔ Popularity	**−0.12**	Very weak — danceability does not predict popularity
Tempo ↔ Energy	**+0.18**	Weak — faster tracks are slightly more energetic on average
Liveness ↔ Popularity	**+0.13**	Weak positive — live recordings show a slight popularity advantage

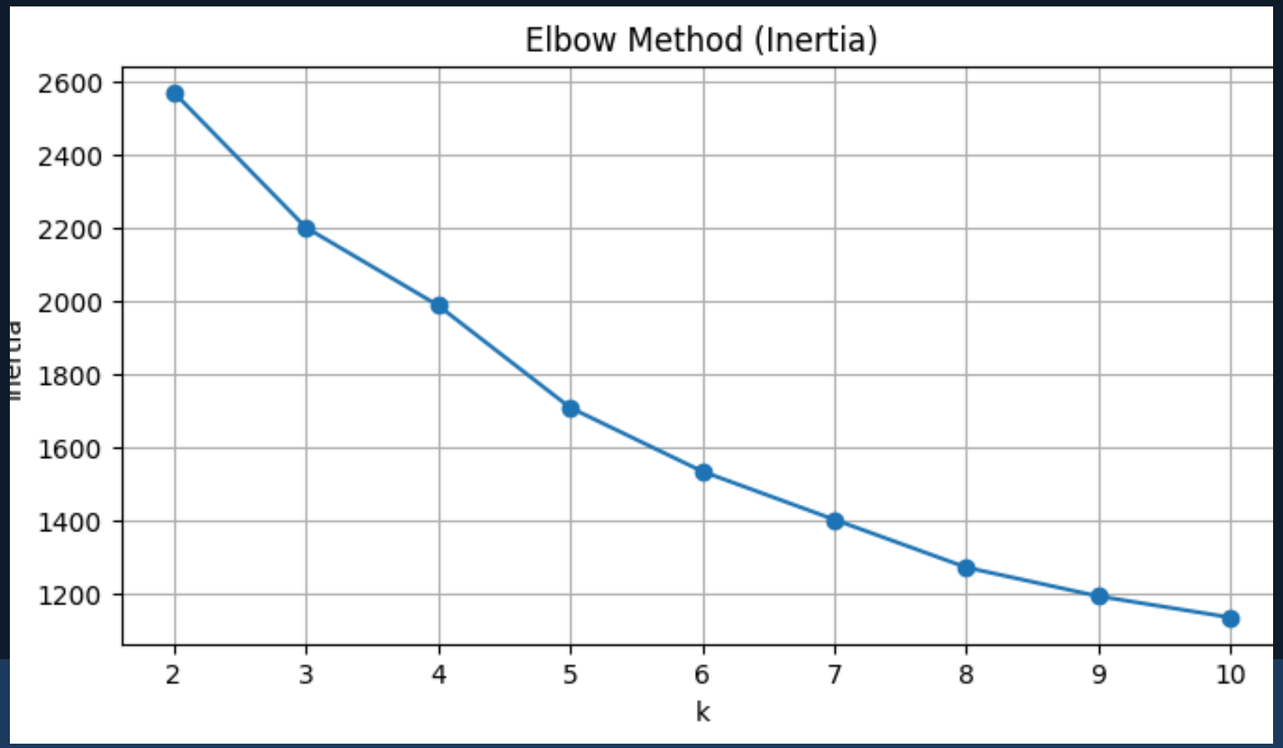
Full Correlation Matrix :

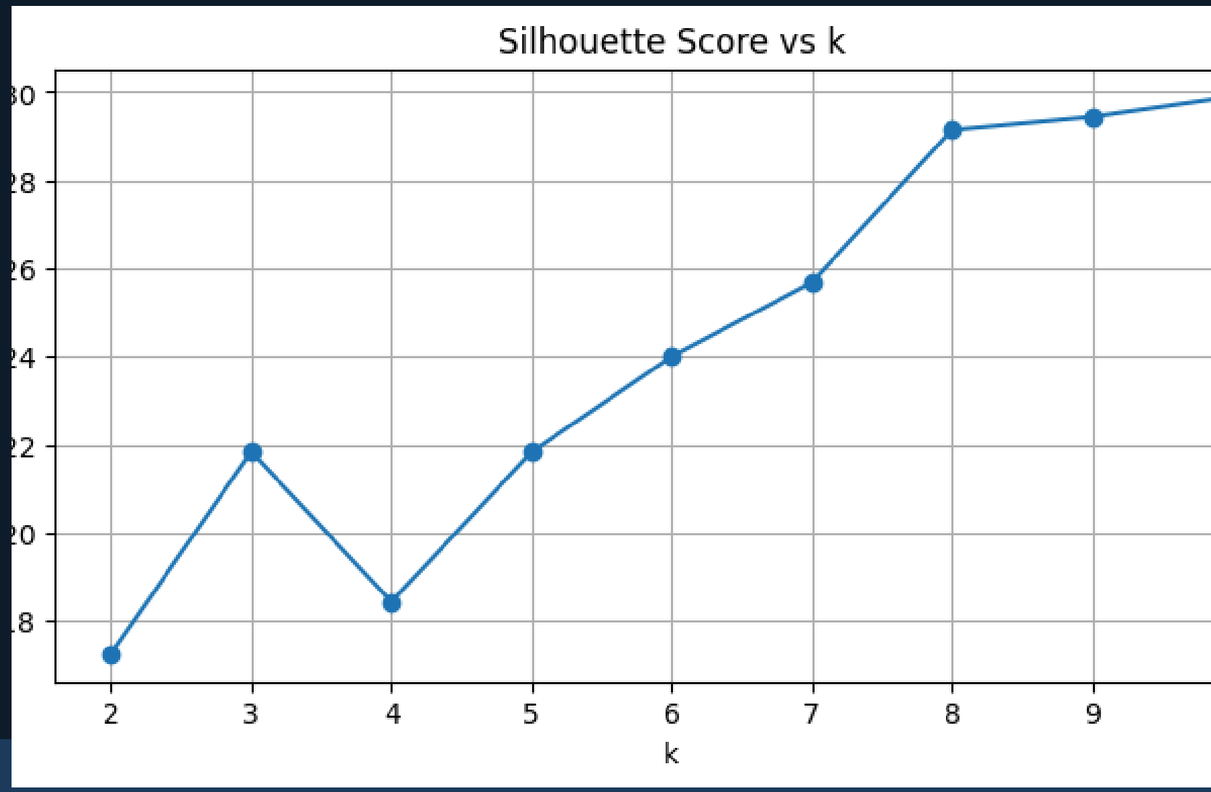


03 — K-Means Clustering (k = 5)

Choosing k = 5

Elbow Method
Inertia drop slows after k ≈ 4–5
Silhouette Analysis
Score peaks at k=10 but k=5 chosen
for interpretability (silhouette ≈ 0.29)





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Algorithm Settings

Init: k-means++ · n_init = 10
random_state = 42 · StandardScaler
Inputs: 13 acoustic features (excl. popularity, explicit)

C0	C1	C2	C3	C4
Danceable / Positive	**Mainstream**	**Acoustic / Relaxed**	**Live / Extended**	**High Intensity**
Dance: 0.75 · Val: 0.69 Tempo: 130 BPM · Pop: 42	Energy: 0.73 · Loud: −5.8dB Pop: 48 (highest)	Acous: 0.55 · Energy: 0.42 Tempo: 105 BPM · Pop: 35	Liveness: 0.56 · 18 min avg Pop: 22 (lowest)	Energy: 0.80 · Tempo: 137 Loud: −4.9dB · Pop: 38

03 — Cluster Centroid Profiles

Feature	C0 Danceable	C1 Mainstream	C2 Acoustic	C3 Live	C4 Intense
Danceability	**0.75 ★**	0.60	0.55	0.52	0.54
Energy	0.70	**0.73 ★**	0.42	0.58	**0.80 ★**
Valence	**0.69 ★**	0.48	0.44	0.40	0.45
Acousticness	0.18	0.17	**0.55 ★**	0.28	0.09
Liveness	0.17	0.18	0.15	**0.56 ★**	0.18
Tempo (BPM)	130	120	105	118	**137 ★**
Loudness (dB)	-6.5	**-5.8 ★**	-10.2	-8.1	**-7.2 ★**
Avg Popularity	42	**48 ★**	35	22	38

04 — DBSCAN Density-Based Clustering

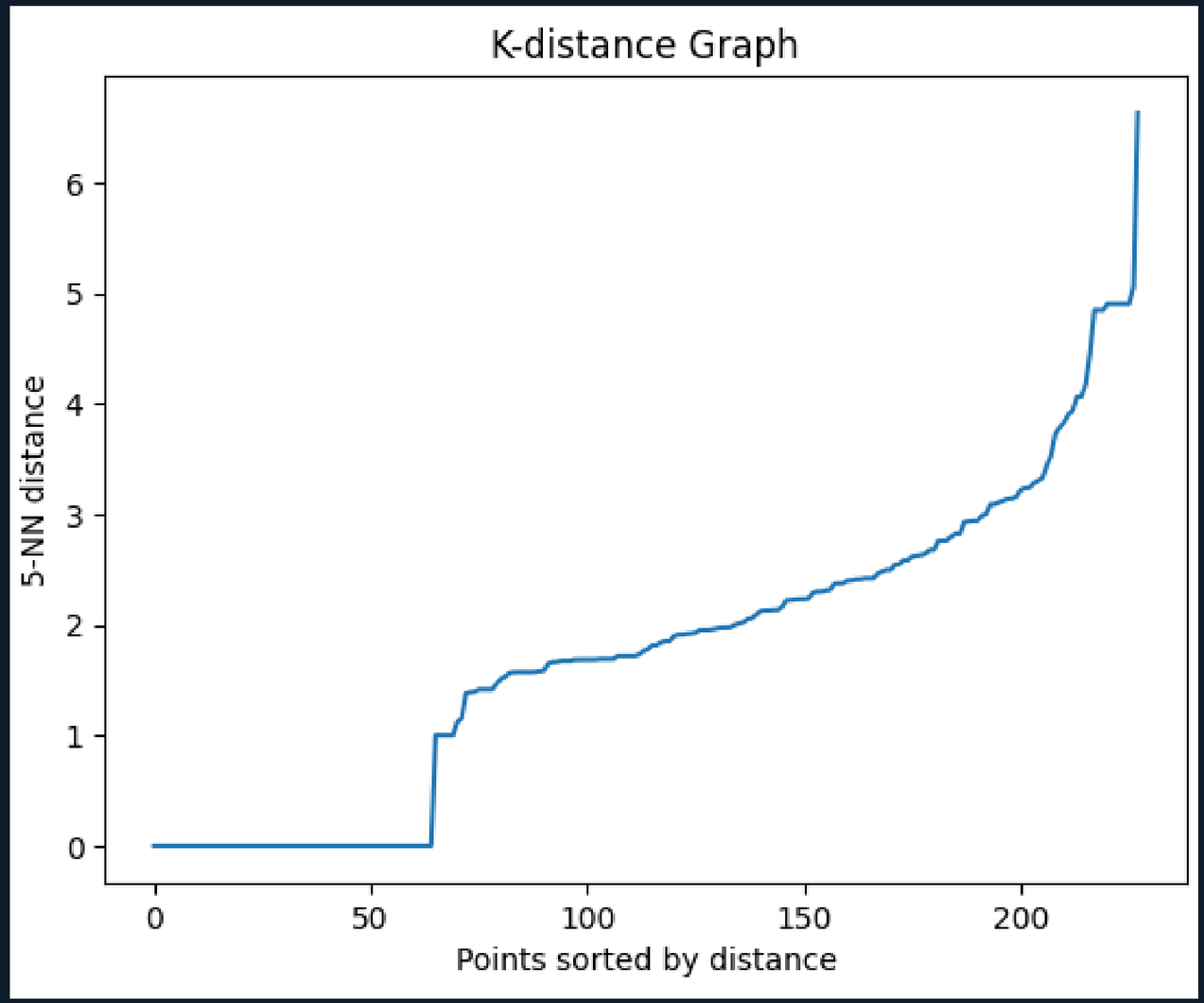
How DBSCAN Works

ε (epsilon)
Neighbourhood radius — controls what is 'nearby'

min_samples
Minimum points to form a dense region

K-Distance Graph
Sort 5-NN distances → find elbow → choose ε

Parameters used:
ε = 2.5 · min_samples = 5



04 — Cluster Comparison

K-Means

cluster	
1	89
0	56
2	50
3	18

DBSCAN

dbscan_cluster	
0	136
-1	45
4	18
3	15
2	7
1	7

K-Means produced five relatively balanced clusters (89, 56, 50, 18, and 15 songs), while DBSCAN identified one large dense cluster along with several smaller clusters and 45 noise points

05 — Dimensionality Reduction

PCA

Principal Component Analysis

Type: Linear

Parameters:

2 components
36.7% variance explained

Key Finding:

Best evaluation scores
Preserves global variance

t-SNE

t-Distributed Stochastic NE

Type: Non-linear

Parameters:

Perplexity = 25
init='pca' · lr='auto'

Key Finding:

Reveals local clusters
Cross-genre emotional groups

ISOMAP

Isometric Mapping

Type: Non-linear (manifold)

Parameters:

k = 8 neighbors
2 components

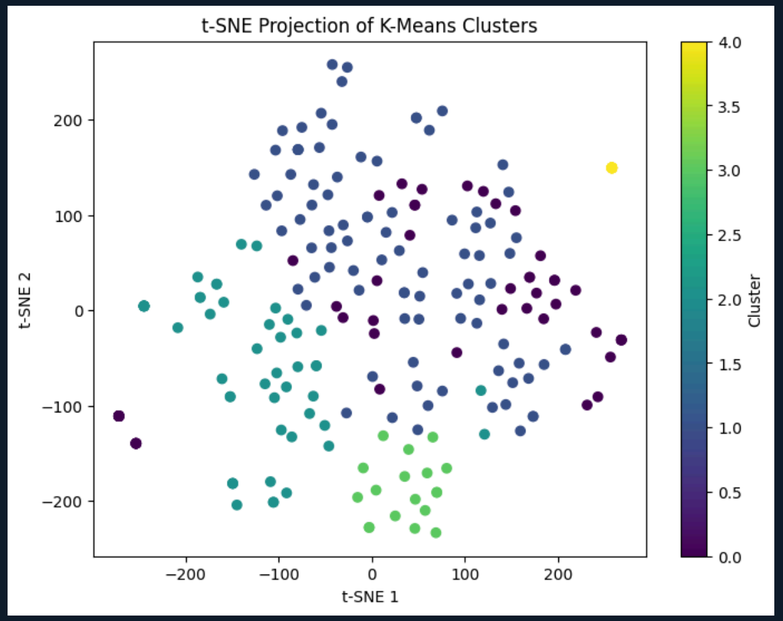
Key Finding:

Acoustic→electronic arc
Geodesic distance preserved

05 — Dimensionality Reduction

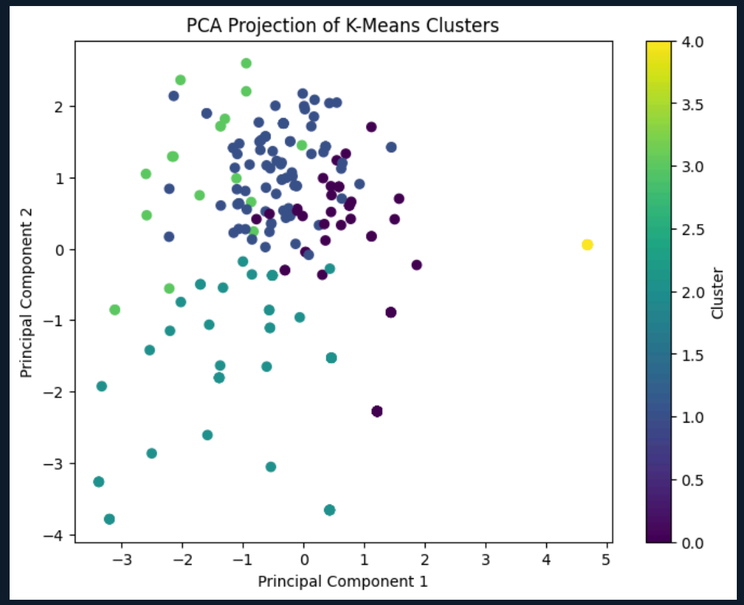
How PCA Works

PCA is a **linear** technique that finds new axes (principal components) capturing the maximum variance in the data. It projects the 13-dimensional acoustic features onto a 2D plane while preserving as much global structure as possible.



How t-SNE Works

t-SNE is a **non-linear** method that emphasizes **local structure**. It maps similar tracks to nearby points in 2D while pushing dissimilar tracks apart, often at the expense of global distances.

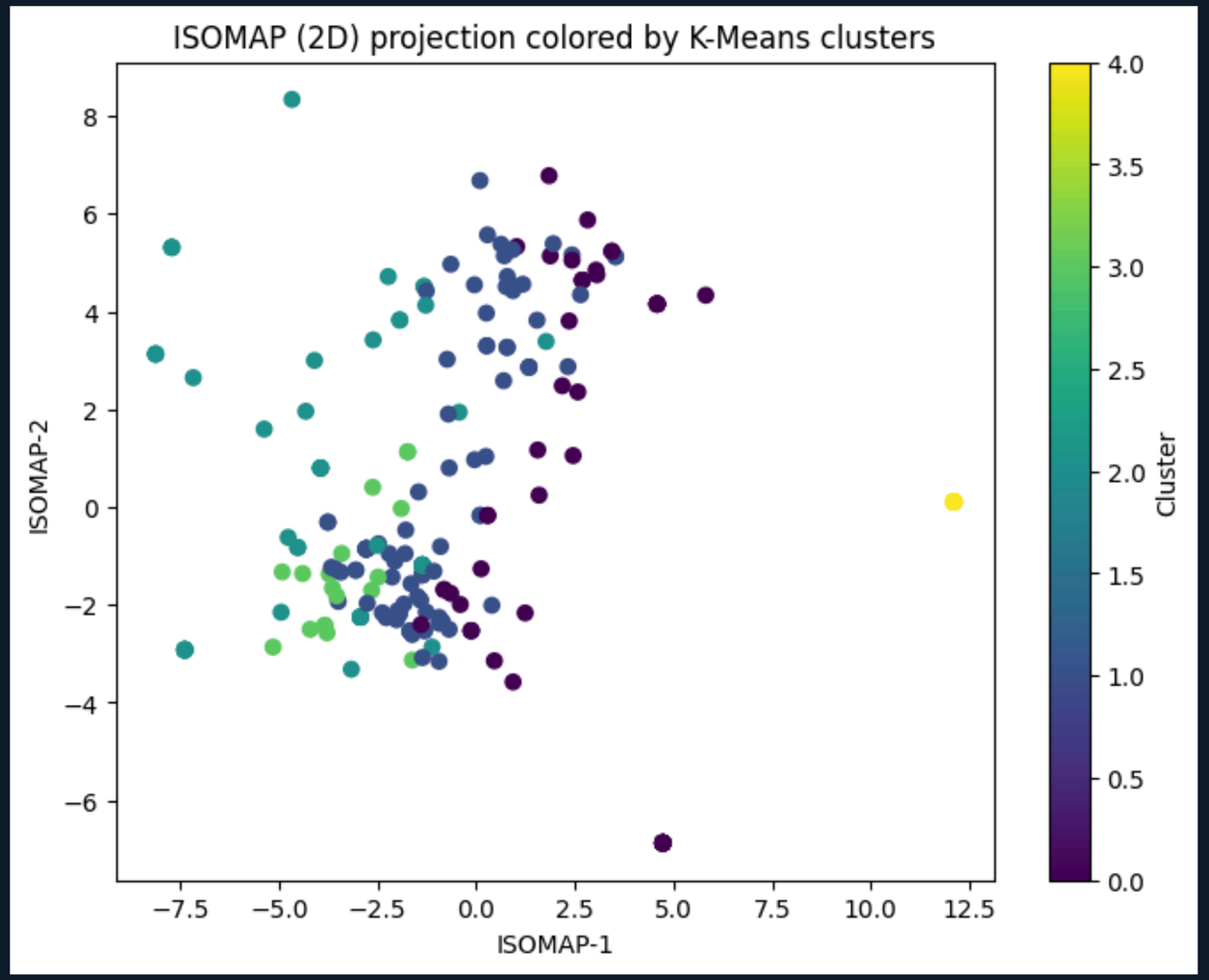


05 — Dimensionality Reduction

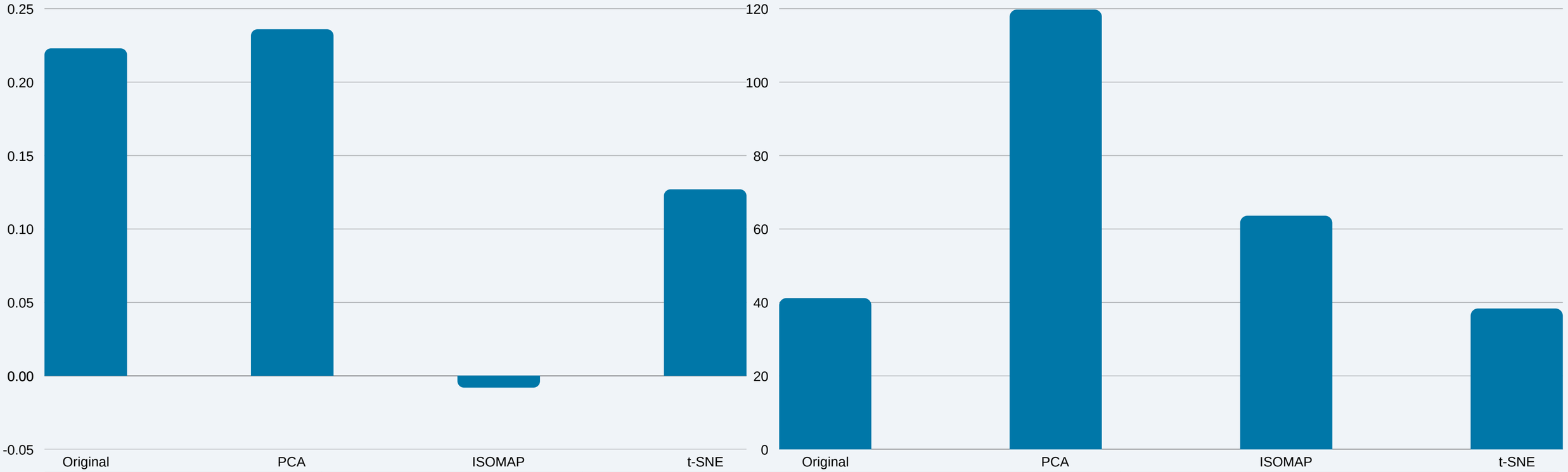
How ISOMAP Works

ISOMAP is a **non-linear manifold learning** technique that estimates **geodesic distances** along the curved surface of the data (like measuring distance on a globe) and then applies classical scaling (MDS) to embed the points in 2D..

- Builds a neighborhood graph (k-nearest neighbors).
- Computes shortest paths between all points (geodesic distances).
- Performs MDS on these distances to find a low-dimensional embedding.



05 — Evaluation Across Feature Spaces



Space	Silhouette ↑	DBI ↓	CHI ↑	Verdict
Original	0.223	1.536	41.19	Baseline
PCA ★	**0.236 ★**	**1.217 ★**	**119.76 ★**	**Best ✓**
ISOMAP	−0.008	1.897	63.62	Visual only

06 — Conclusion

Dataset Fusion

228 enriched tracks built from 2 sources via text-normalized merge on track name + artist.

EDA

Multi-modal popularity distribution. Weak acoustic-popularity correlations (|r| < 0.15). Clustering approach justified.

K-Means (k=5)

5 coherent profiles: Danceable, Mainstream (pop=48), Acoustic, Live/Extended, High-Intensity. Silhouette = 0.223.

DBSCAN

Silhouette = 0.20 — weaker than K-Means. Continuous audio space lacks sharp density boundaries.

Dimensionality Reduction

PCA best preserves clusters (Sil=0.236, DBI=1.22, CHI=119.76). t-SNE & ISOMAP valuable for visualization only.

Future Directions

- Temporal analysis — track cluster evolution over release years
- Supervised extension — cluster labels as features in gradient boosting
- Dataset scaling — 10,000+ tracks for more robust profiles
- Platform expansion — YouTube Music, Tidal, SoundCloud

Thank You



Cross-Platform Music Analytics
Spotify & Apple Music

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